HK IMO Training 2019-2020
Problem Set 15
For 22 February 2020

Problem 57. Let ABC be a triangle with $AB < AC < BC$. Suppose that

- M is the midpoint of $\widehat{AB}$ not containing C on the circumcircle of $\triangle ABC$;
- the line AM meets the line BC at D;
- the line AB meets the circumcircle of $\triangle BMD$ again at F; and
- the line AC meets the circumcircle of $\triangle AMF$ again at E.

Prove that $BD = AE$.

Problem 58. In $\triangle ABC$, the internal angle bisector of $\angle BAC$ meets side BC at D. The lines through D tangents to the circumcircles of $\triangle ABD$ and $\triangle ACD$ meet lines AC and AB at points E and F, respectively. Lines BE and CF intersect at G. Prove that $\angle EDG = \angle ADF$.

Problem 59. An *admissible* expression of a positive integer is an expression as a sum of powers of 3, 4 and 7 in such a way that the expression does not contain two powers with the same base and the same exponent.

For example, $2 = 7^0 + 7^0$ and $22 = 3^2 + 3^2 + 4^1$ are not admissible expressions, while $2 = 3^0 + 7^0$ and $22 = 3^2 + 3^0 + 4^1 + 4^0 + 7^1$ are admissible expressions.

Prove that every positive integer has an admissible expression.

Problem 60. A convex 2020-gon is given for which no three diagonals intersect in a point. The intersection point of two internal diagonals of the given polygon will be called a *knot*. Two knots are called *neighbours* if they share a common diagonal. A closed path between neighbouring knots such that no three consecutive knots share a diagonal will be called a *cycle*.

Find the maximal number of knots that can be coloured such that there exists no coloured cycle, that is, a cycle containing only coloured knots?